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Structural perturbations on huntingtin N17 domain during its folding on 2D-nanomaterials

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Abstract

A globular protein's folded structure in its physiological environment is largely determined by its amino acid sequence. Recently, newly discovered transformer proteins as well as intrinsically disordered proteins may adopt the folding-upon-binding mechanism where their secondary structures are highly dependent on their binding partners. Due to the various applications of nanomaterials in biological sensors and potential wearable devices, it is important to discover possible conformational changes of proteins on nanomaterials. Here, through molecular dynamics simulations, we show that the first 17 residues of the huntingtin protein (HTT-N17) exhibit appreciable differences during its folding on 2D-nanomaterials, such as graphene and MoS₂ nanosheets. Namely, the protein is disordered on the graphene surface but is helical on the MoS₂ surface. Despite that the amphiphilic environment at the nanosheet-water interface promotes the folding of the amphipathic proteins (such as HTT-N17), competitions between protein-nanosheet and intra-protein interactions yield very different protein conformations. Therefore, as engineered binding partners, nanomaterials might significantly affect the structures of adsorbed proteins.

Supplementary material for this article is available [online](#)

Keywords: huntingtin, N17, MoS₂, graphene

(Some figures may appear in colour only in the online journal)

1. Introduction

Proteins inside biological cells operate as nanomachines that perform various biological functions (such as metabolism and signal transduction) to keep the cells viable. The folded structure or conformation of a protein is strongly related to its specific biological functions. Therefore, protein folding remains as an important and challenging research topic, even after decades of experimental and theoretical studies. Taking place in a crowded cell environment, the folding process of a protein into its native structure can be affected/assisted by the

pH condition, ion concentration, temperature and molecular chaperones. Aside from these common factors, it is often perceived that the information for a polypeptide to fold into its native protein structure is mostly encoded in the sequence of amino acids. On the other hand, it is also well known that some so-called transformer proteins can change structures after binding to their host biomolecules, highlighting the importance of binding partners or substrate in protein folding. For example, The C-terminal domain in the RfaH transcription factor undergoes a conformational change from all- β -barrel to all- α -helix after its contact with the N-terminal domain of RfaH [1, 2].

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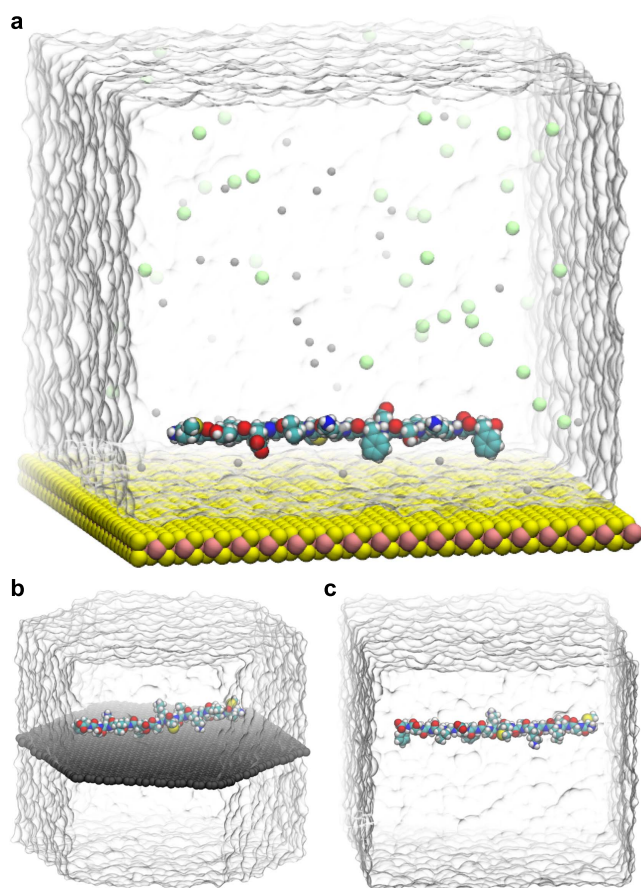


Figure 1. MD simulation systems of a fully extended HTT-N17 near the water-MoS₂ interface (A), near the water-graphene interface (B) and in water (C). Mo, S and C atoms in nanosheets are colored in pink, yellow, and gray, respectively. Na⁺ and Cl[−] in water are colored in green and dark blue, respectively; ions are not shown in (B) and (C). Atoms in HTT-N17 are shown as van der Waals spheres. Water is shown transparently.

Driven by a fast growing interest in bionanotechnologies, nanomedicines and nanosafety, understanding interactions between biomolecules and nanomaterials is urgently demanded. Besides wide applications of graphene and its derivatives as biological sensors (e.g. [3]), monolayers of MoS₂ have been recently applied in field-effect transistors to detect adsorbed proteins [4, 5]. Here, we are motivated to investigate the possible folding/misfolding of a protein interacting with novel engineered nanomaterials, such as a graphene or MoS₂ nanosheet as the binding partner. Both graphene and MoS₂ nanosheets contain hydrophobic surfaces that provide an amphiphilic environment when interfaced with water, similar to many amphiphilic protein/membrane surfaces. Such biomimetic environment can potentially guide or change the folding of amphipathic proteins (or protein segments). For example, it was recently shown that after the adsorption of a protein enzyme on a 2D-material, its α -helix content (in its entire secondary structure) decreased while its β -sheet content increased on the graphene surface [6] or the entire protein can be partially denatured on the MoS₂ surface [7], which further emphasizes the influence of nanomaterials on a protein's

secondary structure and the need to investigate the roles of these novel nanomaterials.

In this proof-of-principle study, we focus on the first 17 residues of the huntingtin (HTT-N17) protein adsorbed on a graphene or MoS₂ nanosheet. An aggregate of HTT proteins can cause the neurodegenerative Huntington's disease that downgrades muscle coordinations and cognitive functions. Despite being unstructured in aqueous solution [8], HTT-N17 (sequence: MATLEKLMKAFESLKSF) can fold into an amphipathic helix at the surface of lipid membranes [9], which may facilitate the protein aggregation and subsequent formation of disease-related fibrils [10]. Additionally, HTT-N17 was found to fold into a helix in the protein-crystal environment [11]. Thus, the secondary structure of HTT-N17 can be highly dependent on its folding environment.

With advances in state-of-the-art supercomputers and ever-improving *insilico* methods [12, 13], molecular dynamics (MD) simulations have been widely used to understand the bio-nano interactions. Particularly for graphene, previous studies have demonstrated graphenes perturbation on protein structures [14–16], graphene's extraction of lipids from a biomembrane [17], graphene's potential interference with protein–protein interactions [18, 19], graphene's enhancement of an enzyme's activity [6], and so on. We recently developed a force field for MoS₂ nanosheets that is compatible with the TIP3P water model [20], which allows us to further study its interactions with biomolecules (such as proteins). Here, using all-atom MD simulations, we investigated possible folding of the amphipathic HTT-N17 on both graphene and MoS₂ nanosheets, which can help understand better the influence of these novel 2D-nanomaterials on protein folding, particularly those amphipathic ones, and also in general help facilitate potential applications of these nanomaterials in biological environment (e.g. biosensors).

2. Simulation methods

Figure 1 illustrates three simulation systems: HTT-N17 at the water-MoS₂ interface (figure 1(A)), HTT-N17 at the water-graphene (figure 1(B)) interface and HTT-N17 in water (figure 1(C)). At the beginning of each simulation system, the HTT-N17 was in a fully extended conformation. The CHARMM22-CMAP force field [21, 22] was applied for the protein. The MoS₂ and graphene nanosheets were rectangular and hexagonal respectively; with periodic boundary conditions, the nanosheets were effectively infinite. The force field for MoS₂ and graphene nanosheets were adopted from previous MD studies [17, 20, 23]. The ion (Na⁺ and Cl[−]) concentration in each system was set at 0.1 M. We used the TIP3P model [24, 25] for water and the standard force field for ions [26]. In total, there are 71 273, 48 003 and 45 691 atoms in simulation systems with HTT-N17 on the MoS₂ nanosheet, on the graphene nanosheet and in water, respectively.

After minimization, each simulation system was equilibrated for 0.5 ns with protein's backbone restrained (with the force constant k at 1 kcal/mol/Å²), in the NPT ensemble

(constant particle number, isobaric and isothermic). This restraint was removed in the following production runs. Atoms (Mo, S or C) in each modeled nanosheet were harmonically restrained (with k at 1 kcal/mol/Å²) to their initial positions in the 2D lattice structure. A smooth cutoff (10–12 Å) was applied to calculate the van der Waals interactions. The electrostatic interactions were calculated using the particle-mesh Ewald method (with the grid size ~ 1 Å). The integration time-step in a simulation was 2 fs, with the SETTLE algorithm [27] to allow rigid bonds. The pressure was maintained at 1 bar using the Nosé–Hoover method [28]. The temperature T was kept at 400 K by applying the Langevin thermostat [29] to all oxygen atoms in water. Here, the elevated temperature facilitates the crossover of energy barriers on the conformational free-energy-landscape, i.e. speeding up the folding process. In simulations with a nanosheet, the cross-section area (or sheet surface area) was constant, by applying the pressure control only in the direction that is normal to the sheet. All MD simulations were carried out using the software package NAMD2.9 [30].

3. Results and discussion

3.1. HTT-N17 is systematically more helical on MoS₂ than in water

In an amphiphilic micelle environment, previous NMR experiments have demonstrated that HTT-N17 forms a helical structure with a disorder coil at one end (PDB ID: 2LD2) as shown in figure 2(A). The other end becomes disordered as well when HTT-N17 is in the 2,2,2-trifluoroethanol solution (PDB ID: 2LD0). Interestingly, HTT-N17 becomes completely helical in the protein crystal environment (PDB ID: 3IO4) [11] but is unstructured in water [8]. Thus, rigorously, the native structure of HTT-N17 is dependent on its folding environment. In this study, for the purpose of discussion/comparison, we artificially define the perfect helix structure (figure 2(B)) as the ‘folded’ (later we refer to as ‘native’) structure of HTT-N17. From the simulation trajectories, we analyzed the time-dependent native contacts of HTT-N17. Optimized for the perfect helix structure, a ‘native’ contact between two nonadjacent i th and j th residues (with $j - i > 2$) is defined if having at least one heavy backbone atom of i th residue within 3.2 Å of any heavy backbone atom in the j th residue, which only includes hydrogen bonds in the helix backbone as ‘native’ contacts. To quantify the folding process, we calculated the time-dependent fraction of ‘native’ contacts $Q = N(t)/N$, where $N(t)$ is the number of native contacts at time t and N is the number of native contacts in the perfect helix structure.

Starting from the fully extended and unfolded structure without any native contact ($N(t = 0) = 0$), the time-dependent Q values for HTT-N17 in each simulation system increased over time as shown in figure 2(C), indicating that the HTT-N17 protein folded gradually. Remarkably, for HTT-N17 on the MoS₂ nanosheet, its Q values increased sharply at around 50 ns, suggesting a protein folding event

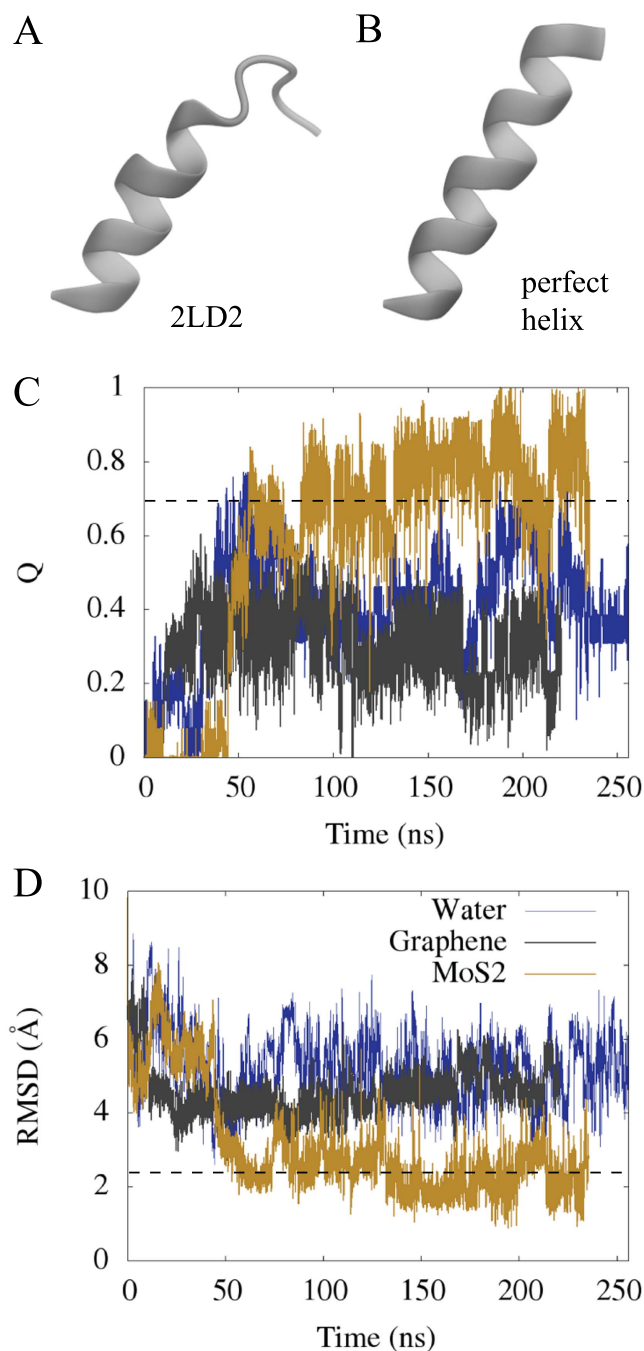


Figure 2. Structure analyses for HTT-N17. (A) The NMR structure of HTT-N17 in micelle (PDB ID: 2LD2). (B) The modeled perfect helical structure for HTT-N17. (C) Q values of HTT-N17 structures (from MD trajectories) compared to the perfect α -helix structure. (D) RMSDs of HTT-N17 structures calculated against the perfect α -helix structure. The data were obtained from the following trajectories: HTT-N17 in water (blue), HTT-N17 on graphene (gray) and HTT-N17 on MoS₂ (gold). Dashed lines show the Q value and the RMSD for the NMR structure computed against the perfect α -helix structure.

(see Movie folding.mpg in supporting information is available online at stacks.iop.org/NANO/28/354001/mmedia). For reference, the Q value for the folded structure in micelle (PDB ID: 2LD2) is 0.7. After the protein folding on the MoS₂ surface, the mean Q value for HTT-N17 is about 0.8. Thus,

the folded structure of HTT-N17 on the MoS₂ surface is even more helical than that in micelle. Noticeably, at certain simulation times, the Q values approached 1, which indicates a perfect helix for HTT-N17. Large variations in Q values result from enhanced thermal fluctuations ($T = 400$ K) in HTT-N17's secondary structure. From the simulation trajectories, we observed that residues at both ends tend to alternate between an unfolded (similar to the one shown in figure 2(A)) and a folded helical structure, yielding fluctuating Q values. Consistently, the root mean square deviations (RMSDs) of the HTT-N17 backbone structure computed against the perfect helix decreased over the simulation time and saturated at around 2 Å. Note that the RMSD of the folded structure in micelle is larger (around 2.57 Å), compared to the perfect helix.

The Q values for HTT-N17 on graphene are significantly smaller and are averaged around 0.3 only. Consistently, from the simulation trajectory, we did not observe any significant α -helical structure of HTT-N17 on graphene. Instead, the HTT-N17 forms a '2D-like helical' structure (see figure 4(C) below) in which some native contacts were preserved. Correspondingly, the saturated RMSD value of HTT-N17 on graphene is around 4.5 Å, much larger than that of HTT-N17 on the MoS₂ surface (figure 2(D)).

As for the Q values of HTT-N17 in water, they fluctuate between 0.2 and 0.7, and are systematically smaller than the ones for HTT-N17 on the MoS₂ surface. The large oscillation in Q values indicate frequent partially folding and unfolding events of the entire HTT-N17 protein in water. Here, besides the force-field related structural variations [31], the elevated temperature (400 K) also promotes the folding of local secondary structures. Note that generally proteins tend to unfold at an even higher temperature in MD simulation [32]. From the simulation trajectory, HTT-N17 in water was most likely to fold temporarily into the helix-coil-helix structure, i.e. the nunchuck-like structure. Because of this deviation from the perfect helix structure, the saturated mean RMSD value is about 5 Å, which is also much larger than that for HTT-N17 on the MoS₂ surface (figure 2(D)).

3.2. Helical residue count of HTT-N17 in different environments

We further studied the time-dependent helicity of HTT-N17 in the three simulation systems. From the simulation trajectories, we calculated the number of helical residues every 20 ps. Generally, the helical residue count of HTT-N17 increases from 0 (fully extended initial structure) to a saturated value after about 50 ns, in all three simulation systems. Averaged from values in the last 100 ns of the simulation, the helical residue count for HTT-N17 in water is 6.77 (± 2.60), which is about 40% of the entire protein (figure 3(A)). Previous MD simulations on HTT-N17 in water with variations of AMBER force field revealed slightly lower percentages (at 300 K) : 36% for the AMBER03 force field, 29.3% for the AMBER99sb*-ILDN force field and 29% for the AMBER99 force field.

Despite the $\sim 30\%$ of native contacts for HTT-N17 on graphene surface (figure 2(C)), the average count for residues

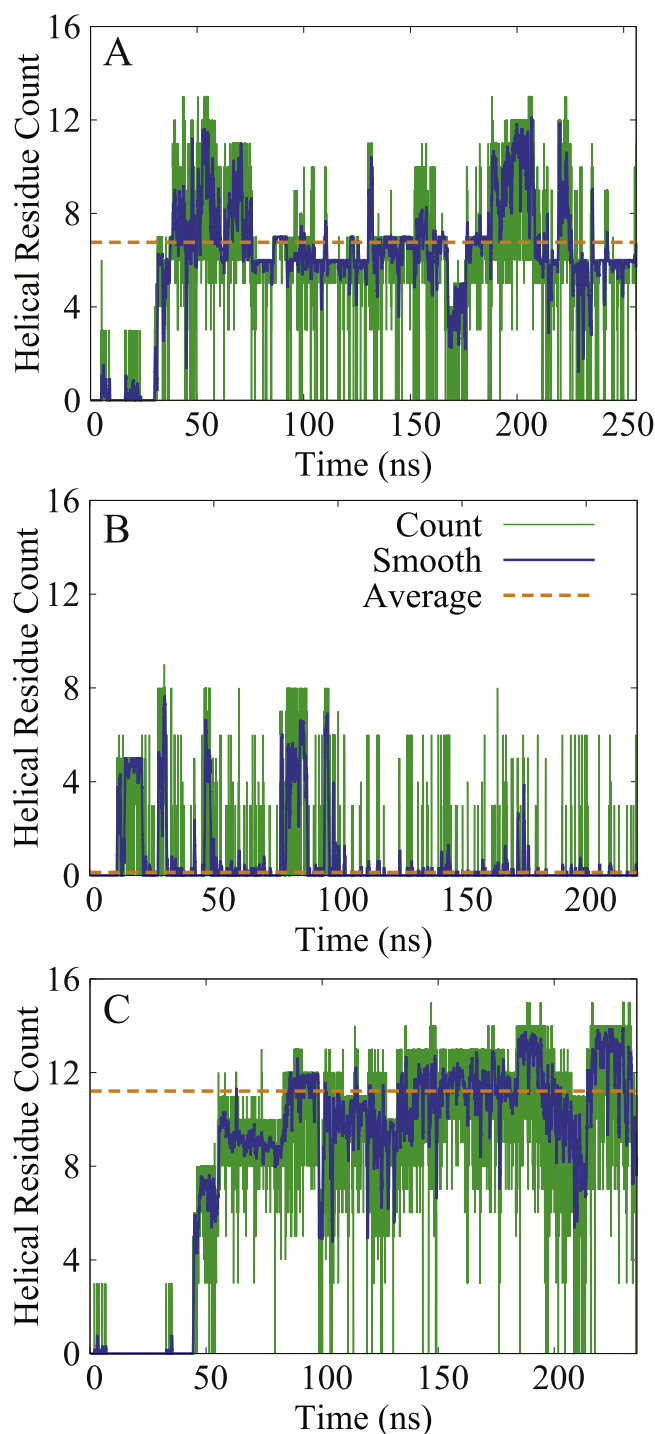


Figure 3. Number of helical residues in HTT-N17 from the following three trajectories: (A) HTT-N17 in water. (B) HTT-N17 on graphene. (C) HTT-N17 on MoS₂. Smoothed curve (blue) is provided on top of the helical residue count (green). The average number of helical residues within the last 100 ns is shown as the orange dashed lines. The residues are defined as helical if the Ψ , Φ angles fall in the range of either α -helix or 3–10 helix calculated by STRIDE [33].

in either α -helix or 3–10 helix conformation is negligible ($\sim 0.13 \pm 0.77$). Thus, HTT-N17 adsorbed on the graphene surface adopts a new flattened '2D-like' conformation (see next section for more details), which is likely resulted from

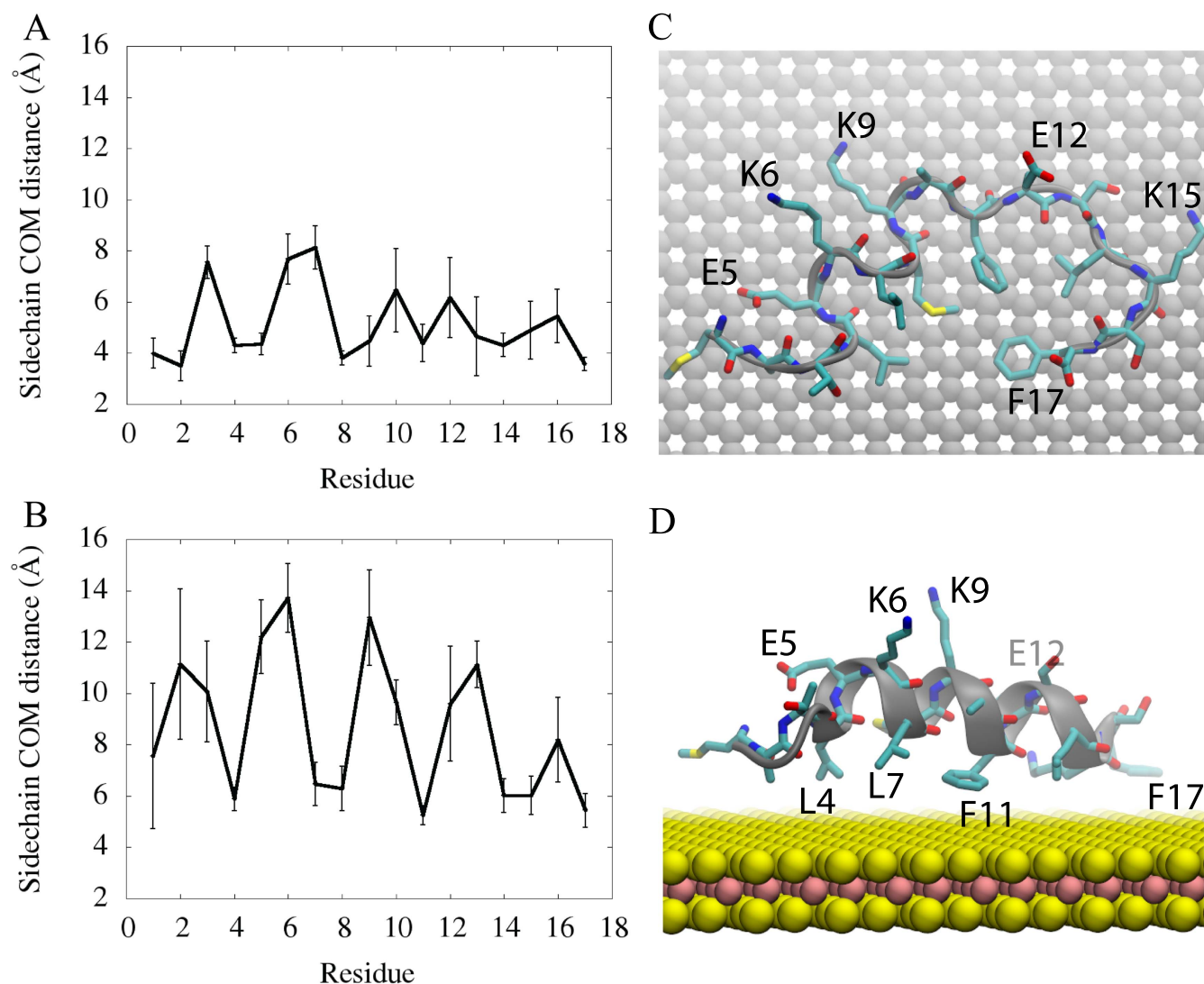


Figure 4. (A) Average COM distances between HTT-N17 side-chains and the graphene sheet. (B) Average COM distances between HTT-N17 side-chains and the S layer (yellow) in the MoS₂ sheet. (C) The top view of HTT-N17 on the graphene sheet. (D) The side view of HTT-N17 on the MoS₂ sheet.

the competition between intra-protein interactions and protein–graphene interactions. Due to the thermal fluctuation/activation, an instant helical residue count can easily reach 6–8, favored by intra-protein interactions. However, protein–graphene interactions dominate most of the times, the resulting helical residue count is close to zero instead.

For HTT-N17 on the MoS₂ surface, the average helical residue count is around 11.21 (± 2.54) that is approximately 66% of the entire protein (figure 3(C)). The maximal value of the helical residue count is 15, which suggests that except for one or two residues of the free N- and/or C-terminals in the 17-residue protein, all other residues can adopt the helical structure. Thus, the HTT-N17 structure is much more helical on MoS₂ than either on graphene or in water. This result is consistent with the large Q values (figure 2(C)) and the small RMSD values (figure 2(D)) shown above.

3.3. The squashed binding structure of HTT-N17 on graphene surface

To highlight the different structures of HTT-N17 binding on graphene and MoS₂, in addition to the backbone structure analyses, we further analyzed the side-chain positions of each residue in HTT-N17 relative to the nanosheet surface. Figures 4(A) and (B) show the center of mass (COM) of each side-chain, averaged from conformations obtained from the simulation trajectories, for HTT-N17 on graphene and MoS₂, respectively. For HTT-N17 on graphene surface, the side-chain COM distances range from 3 to 8 Å and the mean is about 5 Å. By contrast, the side-chain COM distances range from 6 to 13 Å and the mean is about 9 Å, when HTT-N17 binds to the MoS₂ surface, indicating that a side-chain forms a closer contact with the graphene surface than with the MoS₂ surface. For example, the hydrophobic F17 residue lies parallel with both nanosheets, but is about 2 Å closer to the graphene surface. Because of the higher atomic density (and thus the flatness) on the graphene surface, a hydrophobic side

chain can interact more strongly, via the van der Waals (or dispersive) interactions with graphene than with MoS₂. The favorable π - π stacking interactions between F11, F17 of HTT-N17 and graphene surface also contributed to this flattening [34]. Besides the hydrophobic residues, even the hydrophilic and charged residues (e.g. E5, K9 and K15) in HTT-N17 were squeezed onto the graphene surface (figure 4(C)), facilitated by their hydrocarbon chains.

Compared with the graphene surface, the surface of MoS₂ is also hydrophobic with a similar water contact angle [20]. However, due to the low surface atom density on its surface, interactions between the side-chains of HTT-N17 and MoS₂ are relatively weaker, which permit the folding of HTT-N17 on MoS₂. As illustrated in the folded structure (figure 4(D)), hydrophobic residues L4, L7, F11 and F17 are in contact with the hydrophobic MoS₂ surface, while hydrophilic residues E5, K6, K9 and E12 point away from the MoS₂ surface and are surrounded by water molecules. Therefore, the folded HTT-N17 is amphiphilic, which accounts for the observed spontaneous folding at the amphiphilic MoS₂-water interface.

Because of the helical structure, the side-chain COM distances (to the MoS₂ surface) show periodic patterns (figure 4(B)). On the graphene surface, this periodic pattern is significantly less obvious. Specifically, from residue A2 to M8, such pattern is seen with a much reduced amplitude (figure 4(A)). By examining the representative snapshots from the simulation (figure 4(C)), we found that the local helical structure is 'squashed' on the graphene surface, forming a flattened and almost 2D-like structure. Because of this squashed helical structure, few native contacts (hydrogen bonds in the backbone) were preserved, which explains the finite (non-zero) Q values (figure 2(C)). However, its structure deviated from the helical one, which yields large RMSDs (figure 2(D)) and negligible helical residue count (figure 3(B)).

4. Conclusion

In summary, from MD simulations, we observed the folding of a helical structure of HTT-N17 on the MoS₂ surface, but not on the graphene surface, despite their similar hydrophobicity measured by water contact angles, which is found to result from the competition between intra-peptide interactions (mainly hydrogen bonds in the backbone) and side-chains-nanosheet interactions. When intra-protein interactions are dominant, HTT-N17 on the MoS₂ surface folds into a nearly perfect helix. On the other hand, such folding cannot be achieved for HTT-N17 on the graphene surface where sidechains-nanosheet interactions are dominant. This makes the amphiphilic graphene-water interface not suitable for guiding the (amphipathic) protein folding. As for the graphene oxide that contains both hydrophobic sp² domains and hydrophilic oxidized-domains, the folding of HTT-N17 is similar to that on graphene when being adsorbed on a sp² domain but will be quite different on an oxidized domain where the hydrogen-bond interaction may dominate the folding.

HTT-N17 adopts the folding-upon-binding mechanism during its interaction with lipid membranes. The folded amphiphilic helical structure of HTT-N17 possibly further facilitates the targeting or recognizing of endoplasmic reticulum [35] and mitochondria membranes [36]. Our simulation results demonstrate that MoS₂ with weaker interactions with HTT-N17 to some extent mimics the amphiphilic micelle or lipid membrane environment, allowing the folding of HTT-N17. In this regard, MoS₂ could be more biofriendly than graphene. Therefore, potentially, MoS₂ nanosheets may find their applications in novel biosensors and future wearable devices. Our results highlight the possible structural changes of a protein induced by nanomaterials and thus consequent functional variations that can be potentially utilized in many healthcare and industrial applications when considering nanomaterials as binding substrates.

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